Q.29 Explain the working principle of magnetic type speed sensor. (Co4)

Q.30 Explain digital method for torque measurement (CO4)

Q.31 Explain the use of LCD and Seven segment display in measurement system. (CO1)

Q.32 List applications of pneumatic load cell. (CO4)

Q.33 What is capacitive transducer? How it is used to measure linear displacement? (CO2)

Q.34 Write a short note on pH measurement. (CO4)

Q.35 Explain in detail wet and dry bulb thermometer. (CO4)

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

Q.36 Give the working of (CO4)

a) ac tachometer

b) photoelectric speed sensor

Q.37 Explain the working of potentiometer. Give its advantages, disadvantages, and applications. (CO2)

Q.38 Explain the construction and applications of thermistors. Write a short note on positive and negative temperature coefficient materials. (CO4)

Roll No. ....

4th Sem / Elect, Power Station Engg., Elect. & Eltx. Engg.

Subject:- Instrumentation

Time : 3Hrs. M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

Q.1 Transfer function of an instrument is given as _____, where Y=output, x=input (CO1)

a) $Y=f(X^2)$ b) $Y=f(X^3)$

c) $Y=f(X)$ d) none of the above

Q.2 A quartz crystal generates voltage proportional to______ applied upon it. (CO2)

a) Stress b) torque

c) displacement d) all of these

Q.3 LVDT is defined as (CO3)

a) Low Voltage differential transformer

b) Low Voltage displacement transformer

c) Linear voltage differential transformer

d) Linear voltage displacement transformer

Q.4 Stroboscope is used to measure_______ (CO4)

a) displacement b) level

c) thickness d) speed

Q.5 Transducers must have_______ (CO2)

a) High stability b) loading effect

c) sensitivity to unwanted signals

d) frictional error

Q.6 A good strain gauge have______ gauge factor. (CO4)

a) zero b) low

c) one d) high

Q.7 Inductance of inductive transducer is______, L= inductance, S=reluctance (CO4)

a) $L\mu \frac{1}{S}$ b) $L\mu S$

c) $L\mu S^2$ d) Not related at all

Q.8 Piezoelectric transducers are used for the measurement of________ (CO2)

a) Speed b) temperature

c) pressure d) all the above

Q.9 Potentiometer is a _______ transducer (CO2)

a) Passive b) digital

c) active d) inverse

Q.10 Temperature is measured in_______ (CO4)

a) Rankine scale b) Both (a) and (b)

c) Kelvin scale d) None of the above

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

Q.11 Temperature is measured in Kelvin (True/False) (CO4)

Q.12 A method that measures temperature without direct contact is ______. (CO4)

Q.13 Potentiometers are used to measure______. (CO4)

Q.14 What is pH temperature compensation? (CO4)

Q.15 Define measurement? (CO1)

Q.16 Name any negative temperature coefficient material? (CO4)

Q.17 Give applications of electrical transducers. (CO3)

Q.18 Does LVDT provide friction free operation? If yes, explain how? (CO4)

Q.19 What do you mean by signal conditioning? (CO1)

Q.20 On which phenomenon bellows work? (CO4)

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 Name and explain any one of the methods for low pressure measurement. (CO4)

Q.22 Explain any five factors affecting the choice of a transducer. (CO1)

Q.23 Write a short note on force measurement. (CO4)

Q.24 State the applications of pyrometry. (CO4)

Q.25 List the advantages and disadvantages of ultrasonic flow meter. (CO4)

Q.26 Explain the working of strain gauge. (CO4)

Q.27 What do you mean analog transducer and digital transducer? (CO2)

Q.28 Write a short note on Piezoelectric transducer. (CO2)